



Recognising geometric sequences

- The constant multiplier between successive terms in a geometric sequence is called the common ratio. Fill in the blank
- The common ratio of the geometric sequence 7, 28, 112, 448, 1792, ... is 4. Fill in the blank
- 312, 2184, 15 288, 107 016, 749 112, ... is a geometric sequence. Which of these divisions will give you the common ratio? Tick **2** correct answers

☒ $2184 \div 312$

☐ $2184 \div 15288$

☒ $107016 \div 15288$

☐ $107016 \div 749112$

☐ $107016 \div 2184$

- Match each geometric sequence to its common ratio. Write the correct letter in each box

a	16, 80, 400, 2000, ...
b	4.2, 16.8, 67.2, 268.8, ...
c	0.07, 0.56, 4.48, 35.84, ...
d	1602, 4806, 14 418, 43 254, ...
e	20 376, 10 188, 5094, 2547, ...
f	2, 0.4, 0.08, 0.016, ...

e	$\frac{1}{2}$
b	4
f	$\frac{1}{5}$
d	3
c	8
a	5

5 The first term of this geometric sequence is 247. Fill in the blank

n	1 st	2 nd	3 rd	4 th	5 th
T			2223	6669	20 007

6 What could the third term of this geometric sequence be? Tick **2** correct answers

n	1 st	2 nd	3 rd	4 th	5 th
T		10		40	

☒ 20

☐ 25

☒ -20

☐ -25

☐ 80

